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Bridging Literature and Practice: Tool Co-citation Networks from Publications and Registered Workflows

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Introduction

Research software tools, combined in computational workflows, are essential to modern, data-driven, life science. While resources such as InSoLito [REF] provides a network of co-citations of papers associated with software tools, Galaxy [REF] provides a tool recommender based on existing workflows, and Workflowmics [REF] explores new workflows based on annotations in bio.tools [REF], no effort have been so far been made to capture the order of tool citations and compare this with actual, directed, interoperability of tools in workflows. In this project, we bridged ordered citations from the Open Access literature in Europe PMC [REF] with Galaxy, Nextflow [REF] and Snakemake [REF] workflows, using research software annotated and associated with publications in bio.tools to predict tool interoperability from co-citations and visualize relationships between bioinformatics tools used in workflows, focusing on three main goals:

Goal 1: Building directed graph of co-citations of papers associated with software tools in bio.tools. By analyzing the order in which publications associated with bio.tools entries are cited in open-access scientific articles, we infer potential relationships between tools. Unlike conventional undirected co-citation networks, the resulting graph captures citation order and distances, providing information about the sequence in which tools are likely used in computational analyses.

Goal 2: Comparing literature-derived relationships with computational workflows with the latter as ground-truth for the validation of the former. We explore tool dependencies and execution order from workflows deposited in WorkflowHub, including Galaxy and Nextflow workflows, as well as Nextflow workflows in nf-core, and Snakemake workflows from XXX (Julian), and compare these with relationships inferred from the literature. Specifically, we want to know how many co-citations, and with what distance between the citations in the publications, are needed to predict actual execution order in workflows.

Goal 3: Integrating directed tool relationships and visualize them in InSoLiTo. This requires a number of decisions on how to present directionality as well as source of the information shown, e.g. undirected co-citations from all literature, directed co-citations from Open Access subset of the literature, or actual workflows. This will enable users to explore not only which tools are frequently cited together, but also potential directional interoperability between them, providing a foundation for future applications such as software recommendation and automated workflow exploration [REF Workflowmics][REF Pubmetric].

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Methods

Goal 1: Building directed graph of co-citations of papers associated with software tools in bio.tools

Describe methodology here. How the literature was retrieved. Full set only (if we don't show the metabolomics and proteomics specifically). Link to Yojanas/Julians script in GitHub repo.

Network construction in Cytoscape?

Responsible: Yojana, Julian, Magnus?

Goal 2: Comparing literature-derived relationships with computational workflows

Describe methodology here. How the workflows were downloaded from the different sources, and tool edges edges constructed.

Responsible: Alma, Xenia, Julian

Goal 3: Integrating directed tool relationships and visualize them in InSoLiTo.

Responsible: Juancris

Results

Goal 1: Building directed graph of co-citations of papers associated with software tools in bio.tools

Describe results here. Numbers of papers, citations and co-citations of papers in bio.tools, how long it took to download etc. How many nodes and edges in final network? Figure of network in Cytoscape (FIGURE)

Responsible: Yojana, Julian, Magnus?

Goal 2: Comparing literature-derived relationships with computational workflows

Describe results here. How many workflows were retrieved, how many components were matched to bio.tools entries, etc.

Comparison/agreement between co-citations and workflow order (FIGURES)

Responsible: Alma, Xenia, Julian, Vedran

Goal 3: Integrating directed tool relationships and visualize them in InSoLiTo.

How the directed co-citation and workflow interoperability edges were incorporated and visualized in InSoLiTo. By default, we chose to display one edge, with source and target arrowheads, rather than two separate edges, for each source (co-citation, directed co-citation and workflows). Edge widths correspond to the number of co-citation or combination in workflows and relative arrowhead sizes the directedness of the edge.

Responsible: Juancris

Here are some general instructions for this Markdown:

Tables

Tables can be added in the following way, though alternatives are possible:

```
Table: Note that table caption is automatically numbered and should be
given before the table itself.

| Header 1 | Header 2 |
| -------- | -------- |
| item 1 | item 2 |
| item 3 | item 4 |
```

This gives:

Table 1: Note that table caption is automatically numbered and should be given before the table itself.

Header 1	Header 2
item 1	item 2
item 3	item 4

Figures

A figure is added with:

```
![Caption for BioHackrXiv logo figure](./biohackrxiv.png)
```

This gives:



Figure 1: Caption for BioHackrXiv logo figure

Figures can be scaled by adding the width or height to the Markdown like this:

```
![Caption for BioHackrXiv logo figure](./biohackrxiv.png){ width=50px }
```

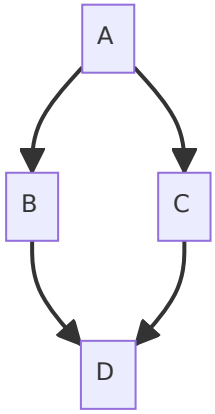
You can add cross references to figures by adding a LaTeX `\label{figureCode}` to the label of the Markdown figure and then use `\ref{figureCode}` to cite it:

```
![Caption for BioHackrXiv logo figure \label{figureCode}](./biohackrxiv.png){ widt
```

This way, we can cite Figure 1.

Mermaid graphs

Mermaid provides a solution to put diagrams in Markdown. BioHackrXiv added support for this in 2026. This is an example of embedding a graph:



Other main section on your manuscript level 1

Lists can be added with:

1. Item 1
2. Item 2

Citation Typing Ontology annotation

You can use CiTO annotations, as explained in this BioHackathon Europe 2021 write up and this CiTO Pilot. Using this template, you can cite an article and indicate *why* you cite that article, for instance DisGeNET-RDF (Queralt-Rosinach et al., 2016).

The syntax in Markdown is as follows: a single intention annotation looks like `[@usesMethodIn:Krewinkel2017]`; two or more intentions are separated with colons, like `[@extends:discusses:Nielsen2017Scholia]`. When you cite two different articles, you use this syntax: `[@citesAsDataSource:Ammar2022ETL; @citesAsDataSource:Arend2022BioHackEU22]`.

Possible CiTO typing annotation include:

- citesAsDataSource: when you point the reader to a source of data which may explain a claim
- usesDataFrom: when you reuse somehow (and elaborate on) the data in the cited entity
- usesMethodIn
- citesAsAuthority
- citesAsEvidence
- citesAsPotentialSolution
- citesAsRecommendedReading
- citesAsRelated
- citesAsSourceDocument
- citesForInformation
- confirms
- documents
- providesDataFor
- obtainsSupportFrom
- discusses
- extends
- agreesWith
- disagreesWith
- updates

There is a general `cites` intention, but this is already implied and should be left out.

Results

Discussion

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Acknowledgements

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References

Author contributions

Alma Nilsson: Methodology, Formal analysis, Visualization, Software, Writing – original draft; Yojana Gadiya: Methodology, Formal analysis, Visualization, Software, Writing – original draft; Julian Uszkoreit: Methodology, Formal analysis, Visualization, Software, Writing – original draft; Juan Crisantos Perez Carrasco: Methodology, Formal analysis, Visualization, Software, Writing – original draft; Magnus Palmblad: Methodology, Formal analysis, Visualization, Software, Writing – original draft.

References

Queralt-Rosinach, N., Pinero, J., Bravo, À., Sanz, F., & Furlong, L. I. (2016). DisGeNET-RDF: harnessing the innovative power of the Semantic Web to explore the genetic basis of diseases. *Bioinformatics*, *32*(14), 2236–2238. https://doi.org/10.1093/bioinformatics/btw214 **[cito:citesAsAuthority]**.

Appendices

If you want the Appendix (-ces) to show up after the references, wrap them in after the header, like done in this Markdown file. Look at the source to see the exact structure.